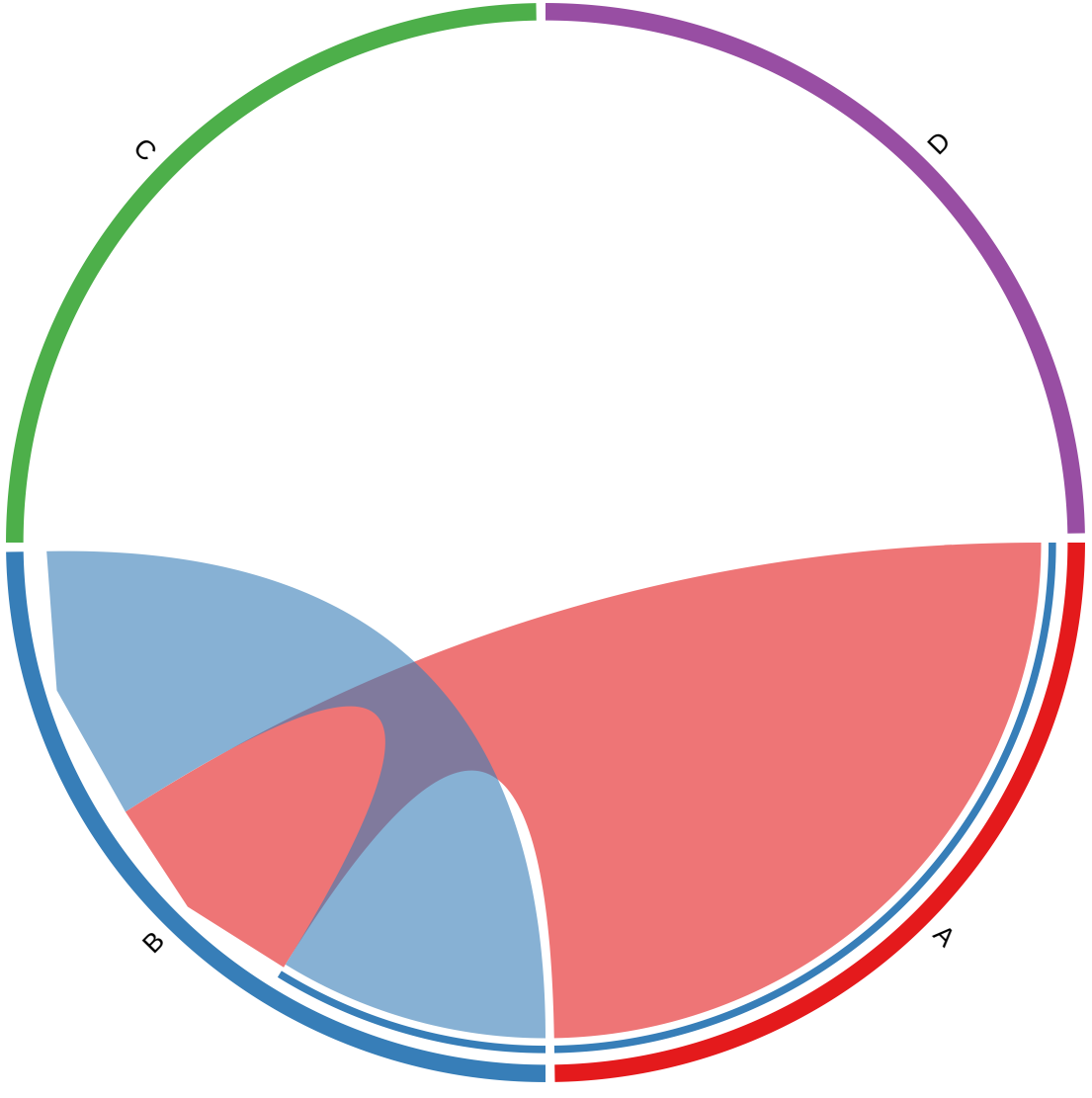P1 signaling pathway network